

A measurable fingerprint of the Voynich manuscript, and a procedure that reproduces it

A computational study of the text of Beinecke MS 408
with three self-corrections and a hand-executable model

Abstract. The text of the Voynich manuscript has resisted reading for a century. This work does not attempt to read it. It measures a fingerprint of forty-four statistics of the text against thirteen natural-language corpora, rules out classes of hypothesis with controlled tests, and demonstrates that the whole fingerprint is reproduced by a procedure containing no language, no cipher and no message — at a mean error of 3.7 % on a held-out half of the corpus (4.3 % on an independent reversed split). The closest prior generator, run here on the same scorecard with the best of twenty of its own configurations, reaches 18.3 %. A reduced, hand-executable version of the model — eight printed tables and a die, no rejection sampling — reaches 7.8 %. Three results are new and are stated precisely: word repetition is scoped to the paragraph and falls to chance across a break; the sections of the book differ partly because the page geometry differs, not only the text; and the headline figure describes the book in aggregate, not any of its parts. Three errors found in earlier revisions of this work are recorded rather than removed.

1. The question

Two claims are often run together. The strong claim is that the text has been read, or can be. The weak and testable claim is that its measurable behaviour is consistent with a procedure that contains no message. Only the second is addressed here, and it is addressed by construction: a model is built that produces the measurements without containing a language, and the standard cryptographic hypotheses for a natural-language plaintext are then excluded by controlled tests.

The manuscript is Yale, Beinecke MS 408, radiocarbon dated 1404–1438. All measurements here are taken from transliterations, principally the Takahashi EVA transcription in IVTFF 2.0, and are therefore statements about the transcription as much as about the ink. Section 13 returns to this.

2. Data and method

The body text — paragraph loci, excluding labels, radial and circular text — comprises 33 641 words in 772 paragraphs over 4 117 lines on 206 folios. Paragraph boundaries are the manuscript's own paragraph marks: a line opening with the transcriber's <%> begins one and a line closing with <\$> ends it. Those two markers occur exactly 772 times each and pair without a single anomaly across the whole file. The locus type is not the paragraph boundary; section 12 records what assuming otherwise cost.

Every result is on a split by leaf parity: the model is fitted on the even-numbered leaves and scored on the odd-numbered ones, or the reverse. All generators are seeded, and the frozen configuration reproduces every published number bit-for-bit (checked drift 0.00 %).

quantity	value
body words	33 641
paragraphs (manuscript marks)	772
lines	4 117
folios	206
mean paragraph length	5.16 lines
mean line length	42.0 characters
comparison corpora	13 (Universal Dependencies)
transliteration alphabets tested	5 (EVA, v101, ZL, FSG, Friedman)

3. The fingerprint

Conditional character entropy — how much the next character carries given the present one — is the statistic that started the field. Bennett reported in 1976 that Voynichese is far below European plain text; the figure here reproduces that and is not new.

corpus	h2 (bits)
Voynich manuscript	1.89
a meaning-free generator (this work)	1.68–1.77
closest of 13 corpora (Augustine)	2.98
Latin / Italian / Spanish	3.19 / 3.19 / 3.19
Old Church Slavonic / Greek / Hebrew	3.49 / 3.58 / 3.85

The gap to the nearest language is 1.09 bits; the gap to the generator is 0.13–0.22 bits. The result survives the obvious objection that EVA writes one manuscript glyph with two Latin characters: across five independent transliteration alphabets, including Currier's v101 where one glyph is exactly one character and the alphabet has 166 symbols, h2 stays between 1.85 and 2.35 while Latin reaches 3.19.

Two further properties are worth recording because the model had to reproduce them, and because they are what a language does not look like.

metric	Voynich	Latin	Italian	OCS	generator
Brevity law (mean length vs rank)	−0.390	−0.927	−1.128	−0.918	−0.429
high-probability bigrams covering the text	21.8 % (9)	1.9 %	1.2 %	1.8 %	22.7 % (8)

In a language, frequent words are short; in the manuscript the effect is almost absent. And a handful of character bigrams accounts for a fifth of all the text.

4. How much context does the text use

Every entropy figure above is a bigram figure. The question that separates a language from a table is the slope: what does a third, fourth or fifth character of context buy? Measured with Witten–Bell smoothing on a held-out half, so the numbers are comparable across alphabets of different size.

corpus	h1	h2	h3	gain 2→3	gain 3→4
manuscript	3.87	2.34	2.14	+0.20	−0.03
generator	3.85	2.33	2.14	+0.19	−0.06
Latin	3.98	3.48	3.03	+0.45	+0.35

The text is exhausted by two characters of context; Latin keeps paying. The generator tracks the manuscript's curve at every order to within 0.01–0.03 bits, and was never fitted to it.

5. What the text is not

A substitution cipher is refuted in the literature, thoroughly, by others. The contribution here is only that the refutation carries a working control, which separates “the attack does not work” from “the hypothesis is false”. The same hill-climbing solver recovers **100 % of words** (6 000 of 6 000) from a random substitution of Latin, and then fails on Voynichese at −11.11 per character against a Latin ceiling of −8.57 and a random-mapping floor of −13.81.

The word-length distribution alone excludes a letter-level cipher over any of the thirteen corpora: the manuscript has 5.8 % words of two letters or fewer where the corpora have 17.9–27.3 %. A transposition or homophonic scheme preserves that distribution and is not excluded by it; it is excluded by the character entropy.

6. The generative model

The model has five components, each entering at a measured property rather than at a guess.

component	what it fixes
BPE syllable inventory (140 merges)	the character-entropy scale
bigram chain over those syllables	the rigid onset and coda slots of a word
P(onset class last character of the previous word)	cross-word dependence
reweighting on (length × final syllable)	the length distribution and the endings at once

A layout layer sits on top: separate line budgets for mid-paragraph and paragraph-final lines, words shortened as the margin approaches, gallows characters at line openings.

The list is not the interesting part. The components are coupled, and most of the obvious ways to fix a single metric break another. Eight dead ends are documented: controlling word length directly destroys the endings (words in *y* fall from 40 % to 23 %); controlling the number of syllables does the same (top-5 finals 91 % → 55 %); appending an ending drawn from the real distribution explodes the vocabulary (TTR 0.28 → 0.43); penalising recent words fixes repeated pairs and flattens the frequency spectrum (top-10 share 11.8 % → 5.5 %). What works is entering at the level where the coupling arises: the identity of the word is never touched, and only conditional distributions are reweighted.

The frozen configuration reproduces the fingerprint at a mean error of **3.7 %** over 44 metrics — 2.7 % across fifteen text statistics and 4.3 % across twenty-nine structural ones — on a held-out half. On the reversed split, where the same configuration is re-run without any selection, the figure is 4.29 %.

tier	metrics	mean error
text statistics	15	2.7 %
structure: lines, paragraphs, Zipf, repetition	29	4.3 %
overall	44	3.7 %
the same, reversed split	44	4.29 %

7. Against the closest prior generator

Timm and Schinner's self-citation generator is the nearest prior work: a complete, public, executable procedure. An earlier revision of this work described its strengths and weaknesses by quoting their paper, which is not a measurement and was also partly wrong. Their executable is now run directly, at a text size matched to the held-out half, and scored on the same vector, with **the best of twenty** configurations of its own switches rather than its default.

generator	tier-1 (15 metrics)	structural (27 metrics)
Timm & Schinner, best of 20 configurations	19.9 %	17.5 %
Timm & Schinner, their default	26.3 %	28.3 %
this model	2.7 %	4.6 %
this model's hand-executable manual	7.3 %	8.4 %

What they have right, measured rather than assumed: mean line length (3 % error), words per line (2 %), words of ten or more characters (6 %), adjacent identical words (6 %), *ch*- onsets (4 %), and all eight positional character entropies within 4–24 %. Their line-fill and gallows machinery works.

Where they diverge is specific, and it is not “vocabulary richness” in general — with the best configuration they over-produce word types, 5 218 against 4 526, while under-concentrating the frequency spectrum. Their output is too even.

metric	held-out	theirs	error
cross-word mutual information	0.068	0.008	88 %
<i>qo</i> - onsets	15.2 %	5.5 %	64 %
words of 8+ characters	8.3 %	12.8 %	54 %
words ending in <i>y</i>	40.3 %	19.1 %	53 %
top-10 word share	12.8 %	7.5 %	42 %
line-length standard deviation	16.05	10.01	38 %
repeated word-pair	15.3 %	10.2 %	33 %

8. The manual: the same fingerprint by hand

The model above reaches 3.7 % by rejection sampling on a joint distribution and by a 151-row transition table. A person with a quill cannot carry either. The question this section answers is the smallest procedure a *hand* can execute, and how close it comes.

The scribe carries eight short tables: sixty syllable shapes; fourteen standard word openings; a twenty-one-row link table giving the shape that usually follows a given letter; a length table; fourteen stock endings chosen by word length; a cross-word carry-over table; a list of habitual word pairs; and page rules. The procedure for one word is: look at the last letter written, write an opening, decide the length and the ending, fill the middle until the word is that size. Nothing is a statistic and nothing is ever thrown away and re-rolled.

	tier-1	tier-2	overall (44 metrics)
computer model, with rejection sampling	2.7 %	4.3 %	3.7 %
the manual, by hand	7.3 %	8.0 %	7.8 %

Four things were learned building it, and each is a constraint on any account of how such a text could be made.

The ending must be carried as letters, not as shapes. A sixty-shape table cannot spell the true last shape of most words, so an ending drawn from the shape table loses the manuscript's endings — *y* above all, which closes 40 % of its words. The output had *y* on 15 % of words until the endings were learned as character suffixes; then 44.6 % against a target of 40.3 %.

The hand's substitute for rejection sampling is deciding the length first. A chain that rolls a die after every shape and stops on a rule gets a geometric length distribution: mean 4.2 against 5.2, standard deviation 2.6 against 1.9, both tails far too heavy. Choosing the size before writing, and filling to it, fixes the distribution outright (sd 1.88 against 1.86) and costs nothing in plausibility — a person does know how long the word is about to be.

Repeats and vocabulary come from one mechanism, and the hand version of it is a swap. The manuscript combines a highly repetitive text with a rich vocabulary; a table-driven writer usually gets one or the other. Both fall out of writing a word again with one shape changed for the one that usually follows it. Restricting that change to a swap — not an insertion or a deletion — matters more than any other single parameter: an insertion also creates repeats, but it doubles the number of one- and two-letter words, because the hand that is supposed to be filling a word to a decided length keeps making it shorter.

What the manual does not reach is vocabulary richness. The residual 7.8 % is dominated by three measures of the same thing: hapax share of tokens 14.7 % against 19.7 %, type-token ratio 0.230 against 0.275, and hapax share of types 63.7 % against 71.6 %. The manuscript's vocabulary is more productive than sixty shapes and one mutation operator produce.

9. The passage is the unit of work

The most informative single measurement in this work concerns word repetition, and it is best stated as a null result. Against its own frequency spectrum — the rate independent sampling from the text's own word distribution would give — the manuscript repeats words at 2.78 times chance *inside* a paragraph, and is **indistinguishable from chance** across a paragraph break.

region	rate	ratio to chance	significance
inside a paragraph	0.907 %	2.78 ×	<i>z</i> = +39.8
across a paragraph break	0.374 %	1.15 ×	<i>z</i> = +1.0
across a folio boundary	0.195 %	0.60 ×	below chance
memoryless baseline (Σf^2)	0.326 %	1.00 ×	—

The property is not that the text repeats more than its spectrum allows. It is that the repetition is scoped exactly to the passage and vanishes at the boundary, while the boundary itself is invisible to every memoryless measure. A language does not do this: topical clustering does not snap to a typographic paragraph mark.

An earlier revision reported that the across-paragraph rate fell *below* chance and that no model in this class reproduced the selectivity. Both were wrong, and the cause was specific: the paragraph pool was the same list for every paragraph, so it raised repetition inside a passage and between passages alike. Giving each passage its own pool — the words already written in that passage, or a sample drawn from the frequency distribution — reproduces the shape, because between two passages the two lists are independent by construction.

configuration	inside	ratio	across	ratio	44-metric error
manuscript	0.907 %	2.78 ×	0.374 %	1.15 ×	—
frozen (no mechanism)	0.267 %	0.88 ×	0.283 %	0.94 ×	3.72 %

sampled pool, 0.80×110	0.875 %	$2.77 \times$	0.321 %	$1.02 \times$	7.56 %
passage self-pool, 0.08	0.664 %	$2.18 \times$	0.218 %	$0.72 \times$	4.94 %
the manual, its own parameters	0.719 %	$2.54 \times$	0.273 %	$0.96 \times$	7.80 %

The last line is the demonstration. The manual carries a passage-scoped pool because a hand naturally does, and its re-use rate was chosen on a selection split for the overall 44-metric error — not for this profile, which had not been measured at that point. It reproduces the shape anyway.

The price of the mechanism in the computer model was four points at first, and almost all of it was one metric: the pool could return the word immediately preceding, which pushed *adjacent identical words* from 21 % to 70 % error. The target profile is measured at lags 2 to 6, so a lag-1 repeat was pure cost with no gain. Excluding the near neighbour from the draw cut the price on the selection split from about 2.5 points to 0.5 and made the profile match better rather than worse.

	within	ratio	across	ratio	44-metric error
frozen configuration	0.260 %	$0.85 \times$	0.304 %	$0.99 \times$	3.96 / 4.30 %
manuscript	0.907 %	$2.78 \times$	0.374 %	$1.15 \times$	—
with the passage pool	0.859 %	$2.85 \times$	0.274 %	$0.91 \times$	4.50 / 6.09 %

The within-paragraph ratio lands on the target on *both* splits, so the mechanism is not fitting the split it was found on. The price is 0.5 points on the selection split and 1.8 on the reversed one, and the reversed split is the honest one; the frozen configuration therefore leaves the mechanism off. That is a documented choice at a measured price rather than a gap.

10. Sections, and what the headline actually measures

Every split in this work is by leaf parity, so both halves carry the same mixture of sections. That makes the 3.7 % a statement about the book in aggregate. Two further measurements establish how far it can be pushed down.

First, the sections differ, and by much more than sampling noise. Measured against a within-section resampling baseline, the ratio of between-section to within-section divergence is $25 \times$ on letter frequencies, $13 \times$ on word onsets, $4.5 \times$ on the first two letters, $3.7 \times$ on word length and $2.8 \times$ on the last two letters. They draw on the same vocabulary — the top twenty word types of every section are in the shared stock, and 60–83 % of every section's running text comes from words occurring in three or more sections — and weight it differently. Part of the difference is not textual at all: mean characters per line runs from 27.0 in the astronomical section to 52.5 in the recipes, tracking how much of the page the drawing occupies. The obvious explanation of Currier's A and B — that the contrast is a gallows-character substitution, *t/k* for *p/f* — is refuted by test: rewriting the herbal with that substitution makes its onset distribution *less* like the recipes, not more (Jensen–Shannon divergence $0.121 \rightarrow 0.160$).

Second, the pooled headline does not survive being taken apart.

design	tier-1	tier-2	overall
A one pooled model, whole held-out half	2.3 %	4.8 %	3.91 %
B the same model, each section alone	9.1–19.1 %	11.4–29.8 %	14.35 %
C one parameter set per section	4.4–9.2 %	6.4–16.8 %	7.50 %

A and B are the same model. The distance between 3.7 % and 14.4 % is not a fitting failure: pooling the sections produces statistics that no individual section has, and the model reproduces the mixture. Per-section parameters halve the per-section error, and their limit is text — the pharmaceutical section has 1 494 training words under that design and lands at 14.2 %, while recipes with 4 982 lands at 6.1 %. A fourth design, a shared text model with per-section page geometry alone, is worse than either at 11.9 %, because the geometry and the length conditioning are coupled.

Which table carries the difference is measurable. Replacing one component of the source model with the target's, one at a time, and measuring how much of the transfer error disappears:

component replaced	mean share of the transfer error removed
the (length $\times$ final shape) joint	22 %
the onset marginal and cross-word table	19 %
the pair memory and stock of frequent words	19 %
line width and paragraph shape	14 %

No component dominates, and no pair falls below about 60 % of its original error even with one table fully replaced. The sections differ in all of their tables at once, which is why per-section parameters are the only thing that works and why they are expensive. The honest

form of the headline is therefore: *a language-free procedure reproduces the aggregate statistics of the manuscript to 3.7 %; applied to any single section it is off by 10–26 %, and with per-section parameters by 5.7–14.2 %.*

11. What remains open

One number is not closed. Within a word, the manuscript retains a small but real character dependence out to a distance of about six characters, and the generator is short of it. The measurement matters because the way it was previously reported was wrong in an instructive way: the repository used a Miller–Madow correction whose magnitude, on a few thousand character pairs against a 22×22 table, was larger than the effect being compared.

Measured instead with a surrogate control — the characters of every word permuted within that word, which preserves length and marginal frequencies exactly and destroys only the dependence, so that the permuted text's mutual information is the estimator's bias — the residual is small and localised.

distance	manuscript (bits)	generator (bits)	gap	in sd
2	0.666	0.652	+0.014	+2.6
3	0.335	0.309	+0.026	+5.8
4	0.097	0.075	+0.022	+4.1
5	0.075	0.055	+0.021	+1.7
6	0.077	0.044	+0.033	+2.5
8	−0.026	−0.008	—	both zero

Beyond a distance of six neither corpus has any measurable dependence: the corrected value is at or below zero for both. The residual is 0.022–0.032 bits over distances three to six, significant at 5.8 and 4.1 standard deviations at distances three and four.

Labelling every character pair by where it sits in the word locates the deficit rather than spreading it. Eighty-six per cent of it is in the *front* half of the word; the middle is under one standard deviation at every distance; and at the back the generator has *more* dependence than the manuscript. The sharpest single cell is pairs starting at position 0 or 1, at +7.4 standard deviations at distance 3.

d	manuscript	chain alone	chain + reweighting	frozen model
3	0.201	0.221	0.178	0.162
4	0.041	0.109	0.014	0.017
5	0.016	0.007	0.011	0.004

Front half of the word only, so this is not a length effect. The chain on its own puts more dependence there than the manuscript has; the (length $\times$ ending) reweighting on top of it removes more than it should; the frozen model lands below the target.

Four attempts were made to close it, and the fourth is the informative one.

attempt	result
a within-word repetition mechanism	no effect
conditioning the second shape on the onset class	helps the split it was found on, a wash on the reversed one
<i>p_raw</i> : accept a fraction of words unchecked	reaches the target only by wrecking the length tails (<i>len</i> ≥10 5 % → 22 %, <i>len</i> ≤2 3 % → 15 %)
<i>endfix</i> : accept on length alone, then repair the ending	much worse — front-half MI 0.154 → 0.079, 44-metric error 3.96 % → 11.97 %

The last failure is the informative one. The idea was to decouple the two halves of the reweighting, since the front is what the acceptance damages. It fails because the ending is not an appendage: it is constrained by what precedes it, and editing it independently destroys the dependence it was meant to preserve. A word-level mixture of checked and unchecked words *does* reach the target, at 25 % unchecked and a small cost in the length distribution — the interpolation exists, and the model's single joint accept/reject cannot express it.

Conclusion. The front-of-word dependence is not reachable within this architecture. The reweighting is one accept-or-reject on (length, ending); the manuscript's value sits between what the chain proposes and what that acceptance keeps, and every attempt to split the acceptance apart makes it worse. Closing it would need a different generator rather than a different parameter. This is the one item in the work that is not closed, and it is a limit of the model rather than a missing measurement.

12. Three corrections

Recorded because each changed a published figure, and because the third is a trap that any study of this transcription could fall into.

1. A leaked layout parameter. Line and paragraph constants were once taken from the held-out document. That is a leak, it inflated the result, and it was found only because a reversed split was run. The honest within-section error moved from 3.2 % to 4.0 %.

2. A mutual-information comparison measuring its own estimator bias. The Miller–Madow correction on the table sizes involved, 0.168 bits, was larger than the effect being compared. Corrected, the manuscript and generator agree to 0.0016 bits where a gap had been reported.

3. Paragraph segmentation by the wrong marker. Paragraphs were delimited by the IVTFF locus type (@ / + / *) instead of by the manuscript's own paragraph marks (<%> / <\$>). The @ marker appears only on the first paragraph of a page, so in the recipes section this merged about twelve real paragraphs into one 43-line block. Real mean paragraph length is **5.16 lines, not 14.81**. Every paragraph-dependent number was recomputed and the frozen configuration re-verified; the headline moved from 4.0 % to 3.7 %, improving, because the generator had been faithfully learning the artefact. A claim about repetition reversing sign across a paragraph break was withdrawn as a consequence.

13. What is claimed

Claimed. The measurable behaviour of the text — forty-four statistics covering character entropy and its context profile, vocabulary richness, the word-frequency spectrum, length and ending distributions, onset classes, cross-word dependence, line fill, paragraph shape, gallows placement, Zipf slope and repetition structure — is reproduced by a procedure that contains no language, no cipher and no message (3.7 %); a reduced version of that procedure is executable by hand with a printed table and a die (7.8 %); and the standard cryptographic hypotheses for a natural-language plaintext are excluded by controlled tests.

Not claimed. That the manuscript is meaningless. A verbose cipher or a nomenclator would be statistically indistinguishable from the generator's output. The defensible statement is that the data are inconsistent with a natural language under any letter-level cipher and consistent with procedural generation. Nothing here bears on who wrote the manuscript, when, or whether every stroke is original; those are separate literatures.

Scope. Every number is a statement about a transliteration, not about the ink. The same statistics were computed across five independent transliteration alphabets and the qualitative results survive, but the objection is real and is answered empirically rather than in principle.

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All figures in this document are produced by the code in `analysis/` and are reproducible from a clean checkout; the frozen configuration verifies bit-for-bit at 0.00 % drift. Three negative results are kept in the repository rather than removed: an image-based test of the writing direction that failed on two of three pages, a compression test that does not discriminate, and an aborted reversal test whose statistic is invariant under reversal and therefore could not have answered its question.